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Navneet_1458_Day-2

```
%CSE General-L
% https://in.mathworks.com/help/matlab/language-fundamentals.html
```

```
clc
clear all
close all
```

Use of "..."

```
clc
a= 1 - 1/2 + - 1/3 ...
    - 1/4 + - 1/5
mytext = ['asdfhj'...
          'dfghjk']
a='123'
length(a)
exist fprintf
```

```
a =

    -0.2833
```

```
mytext =

    'asdfhjdfghjk'
```

```
a =

    '123'
```

```
ans =  
  
    3
```

```
ans =  
  
    5
```

if-else looping

```
A=10  
b=20  
if A>b  
    "Greter"  
elseif A<b  
    "Less"  
end
```

```
A =  
  
    10
```

```
b =  
  
    20
```

```
ans =  
  
    "Less"
```

Matrix and functions

```
round(100*rand(1,100))  
A=true(4,3)  
false(1,2)  
tm=magic(5)  
tmq=tm(1:2,4:5)  
A=logical(sparse(5,3));  
whos A
```

```
ans =  
  
    Columns 1 through 13
```

99	73	30	34	73	99	70	9	47	64	53	14	28
----	----	----	----	----	----	----	---	----	----	----	----	----

Columns 14 through 26

29	79	93	35	21	68	36	44	91	83	39	73	29
----	----	----	----	----	----	----	----	----	----	----	----	----

Columns 27 through 39

62	30	2	92	69	23	50	79	38	22	72	77	69
----	----	---	----	----	----	----	----	----	----	----	----	----

Columns 40 through 52

44	76	51	33	0	58	17	19	15	51	2	47	5
----	----	----	----	---	----	----	----	----	----	---	----	---

Columns 53 through 65

36	16	57	82	9	23	25	35	41	11	41	68	54
----	----	----	----	---	----	----	----	----	----	----	----	----

Columns 66 through 78

63	66	68	13	44	94	57	58	76	25	69	29	57
----	----	----	----	----	----	----	----	----	----	----	----	----

Columns 79 through 91

64	40	25	94	45	53	63	67	44	17	69	92	37
----	----	----	----	----	----	----	----	----	----	----	----	----

Columns 92 through 100

89	48	63	47	11	11	11	45	94
----	----	----	----	----	----	----	----	----

A =

4×3 logical array

1	1	1
1	1	1
1	1	1
1	1	1

ans =

1×2 logical array

0	0
---	---

tm =

17	24	1	8	15
23	5	7	14	16
4	6	13	20	22
10	12	19	21	3
11	18	25	2	9

tmq =

8	15
14	16

<i>Name</i>	<i>Size</i>	<i>Bytes</i>	<i>Class</i>	<i>Attributes</i>
<i>A</i>	5x3	41	<i>logical</i>	<i>sparse</i>

Grids

a=1,2,3,4
b=1,2,3
y=logspace(a,b)

a =

1

ans =

2

ans =

3

ans =

4

b =

1

ans =

```

2

ans =

3

y =

Columns 1 through 13
10    10    10    10    10    10    10    10    10    10    10    10    10

Columns 14 through 26
10    10    10    10    10    10    10    10    10    10    10    10    10

Columns 27 through 39
10    10    10    10    10    10    10    10    10    10    10    10    10

Columns 40 through 50
10    10    10    10    10    10    10    10    10    10    10

%Frequency Space
m=2
n=3
[f1,f2] = freqspace(n)
[f1,f2] = freqspace([m n])
[x,y] = freqspace('meshgrid')
N=6
f = freqspace(N)

f = freqspace(N, 'whole')
ndims(A)
numel(f)
isscalar(A)
a=[1 2 3;1 2 3]
isvector(a)
ismatrix(a)
A3 = zeros(1,3,2)
ismatrix(A3)
aa=[1,2,3;1,2,3]
isrow(aa)
isrow(aa')

```

$m =$

2

$n =$

3

$f1 =$

-0.6667 0 0.6667

$f2 =$

-0.6667 0 0.6667

$f1 =$

-0.6667 0 0.6667

$f2 =$

-1 0

$x =$

Columns 1 through 7

-0.9901 -0.9703 -0.9505 -0.9307 -0.9109 -0.8911 -0.8713

Columns 8 through 14

-0.8515 -0.8317 -0.8119 -0.7921 -0.7723 -0.7525 -0.7327

Columns 15 through 21

-0.7129 -0.6931 -0.6733 -0.6535 -0.6337 -0.6139 -0.5941

Columns 22 through 28

-0.5743 -0.5545 -0.5347 -0.5149 -0.4950 -0.4752 -0.4554

Columns 29 through 35

-0.4356 -0.4158 -0.3960 -0.3762 -0.3564 -0.3366 -0.3168

Columns 36 through 42

-0.2970	-0.2772	-0.2574	-0.2376	-0.2178	-0.1980	-0.1782
---------	---------	---------	---------	---------	---------	---------

Columns 43 through 49

-0.1584	-0.1386	-0.1188	-0.0990	-0.0792	-0.0594	-0.0396
---------	---------	---------	---------	---------	---------	---------

Columns 50 through 56

-0.0198	0	0.0198	0.0396	0.0594	0.0792	0.0990
---------	---	--------	--------	--------	--------	--------

Columns 57 through 63

0.1188	0.1386	0.1584	0.1782	0.1980	0.2178	0.2376
--------	--------	--------	--------	--------	--------	--------

Columns 64 through 70

0.2574	0.2772	0.2970	0.3168	0.3366	0.3564	0.3762
--------	--------	--------	--------	--------	--------	--------

Columns 71 through 77

0.3960	0.4158	0.4356	0.4554	0.4752	0.4950	0.5149
--------	--------	--------	--------	--------	--------	--------

Columns 78 through 84

0.5347	0.5545	0.5743	0.5941	0.6139	0.6337	0.6535
--------	--------	--------	--------	--------	--------	--------

Columns 85 through 91

0.6733	0.6931	0.7129	0.7327	0.7525	0.7723	0.7921
--------	--------	--------	--------	--------	--------	--------

Columns 92 through 98

0.8119	0.8317	0.8515	0.8713	0.8911	0.9109	0.9307
--------	--------	--------	--------	--------	--------	--------

Columns 99 through 101

0.9505	0.9703	0.9901
--------	--------	--------

$y =$

Columns 1 through 7

-0.9908	-0.9725	-0.9541	-0.9358	-0.9174	-0.8991	-0.8807
---------	---------	---------	---------	---------	---------	---------

Columns 8 through 14

-0.8624	-0.8440	-0.8257	-0.8073	-0.7890	-0.7706	-0.7523
---------	---------	---------	---------	---------	---------	---------

Columns 15 through 21

-0.7339	-0.7156	-0.6972	-0.6789	-0.6606	-0.6422	-0.6239
---------	---------	---------	---------	---------	---------	---------

Columns 22 through 28

-0.6055 -0.5872 -0.5688 -0.5505 -0.5321 -0.5138 -0.4954

Columns 29 through 35

-0.4771 -0.4587 -0.4404 -0.4220 -0.4037 -0.3853 -0.3670

Columns 36 through 42

-0.3486 -0.3303 -0.3119 -0.2936 -0.2752 -0.2569 -0.2385

Columns 43 through 49

-0.2202 -0.2018 -0.1835 -0.1651 -0.1468 -0.1284 -0.1101

Columns 50 through 56

-0.0917 -0.0734 -0.0550 -0.0367 -0.0183 0 0.0183

Columns 57 through 63

0.0367 0.0550 0.0734 0.0917 0.1101 0.1284 0.1468

Columns 64 through 70

0.1651 0.1835 0.2018 0.2202 0.2385 0.2569 0.2752

Columns 71 through 77

0.2936 0.3119 0.3303 0.3486 0.3670 0.3853 0.4037

Columns 78 through 84

0.4220 0.4404 0.4587 0.4771 0.4954 0.5138 0.5321

Columns 85 through 91

0.5505 0.5688 0.5872 0.6055 0.6239 0.6422 0.6606

Columns 92 through 98

0.6789 0.6972 0.7156 0.7339 0.7523 0.7706 0.7890

Columns 99 through 105

0.8073 0.8257 0.8440 0.8624 0.8807 0.8991 0.9174

Columns 106 through 109

0.9358 0.9541 0.9725 0.9908

N =

```

        6

f =
      0      0.3333      0.6667      1.0000

f =
      0      0.3333      0.6667      1.0000      1.3333      1.6667

ans =
      2

ans =
      6

ans =
    logical
      0

a =
      1      2      3
      1      2      3

ans =
    logical
      0

ans =
    logical
      1

A3(:, :, 1) =
      0      0      0

```

```
A3(:, :, 2) =  
  
    0    0    0
```

```
ans =  
  
logical  
  
0
```

```
aa =  
  
    1    2    3  
    1    2    3
```

```
ans =  
  
logical  
  
0
```

```
ans =  
  
logical  
  
0
```

Head/Tail Functions

```
A=magic(16)  
x=A'  
(head(x,3))'  
tail(A)  
A=[1,2,3,4]  
b1=resize(A,2)  
b1=resize(A,6)  
  
A=1:10;  
B=reshape(A,[5,2])  
B = permute(A,[3 2 1])
```

```
z=magic(4)  
a=(sort(z'))'  
sort(a)  
whos("a")
```

A =

Columns 1 through 13

256	2	3	253	252	6	7	249	248	10	11	245	
244	17	239	238	20	21	235	234	24	25	231	230	28
29	33	223	222	36	37	219	218	40	41	215	214	44
45	208	50	51	205	204	54	55	201	200	58	59	197
196	192	66	67	189	188	70	71	185	184	74	75	181
180	81	175	174	84	85	171	170	88	89	167	166	92
93	97	159	158	100	101	155	154	104	105	151	150	108
109	144	114	115	141	140	118	119	137	136	122	123	133
132	128	130	131	125	124	134	135	121	120	138	139	117
116	145	111	110	148	149	107	106	152	153	103	102	156
157	161	95	94	164	165	91	90	168	169	87	86	172
173	80	178	179	77	76	182	183	73	72	186	187	69
68	64	194	195	61	60	198	199	57	56	202	203	53
52	209	47	46	212	213	43	42	216	217	39	38	220
221	225	31	30	228	229	27	26	232	233	23	22	236
237	16	242	243	13	12	246	247	9	8	250	251	5
4												

Columns 14 through 16

14	15	241
227	226	32
211	210	48
62	63	193
78	79	177
163	162	96
147	146	112
126	127	129
142	143	113
99	98	160
83	82	176
190	191	65
206	207	49
35	34	224

19	18	240
254	255	1

x =

Columns 1 through 13

256	17	33	208	192	81	97	144	128	145	161	80	
64	2	239	223	50	66	175	159	114	130	111	95	178
194	3	238	222	51	67	174	158	115	131	110	94	179
195	253	20	36	205	189	84	100	141	125	148	164	77
61	252	21	37	204	188	85	101	140	124	149	165	76
60	6	235	219	54	70	171	155	118	134	107	91	182
198	7	234	218	55	71	170	154	119	135	106	90	183
199	249	24	40	201	185	88	104	137	121	152	168	73
57	248	25	41	200	184	89	105	136	120	153	169	72
56	10	231	215	58	74	167	151	122	138	103	87	186
202	11	230	214	59	75	166	150	123	139	102	86	187
203	245	28	44	197	181	92	108	133	117	156	172	69
53	244	29	45	196	180	93	109	132	116	157	173	68
52	14	227	211	62	78	163	147	126	142	99	83	190
206	15	226	210	63	79	162	146	127	143	98	82	191
207	241	32	48	193	177	96	112	129	113	160	176	65
49												

Columns 14 through 16

209	225	16
47	31	242
46	30	243
212	228	13
213	229	12
43	27	246
42	26	247
216	232	9
217	233	8
39	23	250
38	22	251

220	236	5
221	237	4
35	19	254
34	18	255
224	240	1

ans =

256	2	3
17	239	238
33	223	222
208	50	51
192	66	67
81	175	174
97	159	158
144	114	115
128	130	131
145	111	110
161	95	94
80	178	179
64	194	195
209	47	46
225	31	30
16	242	243

Columns 1 through 13

128	130	131	125	124	134	135	121	120	138	139	117	
116	145	111	110	148	149	107	106	152	153	103	102	156
157	161	95	94	164	165	91	90	168	169	87	86	172
173	80	178	179	77	76	182	183	73	72	186	187	69
68	64	194	195	61	60	198	199	57	56	202	203	53
52	209	47	46	212	213	43	42	216	217	39	38	220
221	225	31	30	228	229	27	26	232	233	23	22	236
237	16	242	243	13	12	246	247	9	8	250	251	5
4												

Columns 14 through 16

142	143	113
99	98	160
83	82	176
190	191	65
206	207	49
35	34	224
19	18	240

```

254  255    1

A =

    1    2    3    4

b1 =

    1    2

b1 =

    1    2    3    4    0    0

B =

    1    6
    2    7
    3    8
    4    9
    5   10

B =

    1    2    3    4    5    6    7    8    9   10

z =

   16    2    3   13
    5   11   10    8
    9    7    6   12
    4   14   15    1

a =

    2    3   13   16
    5    8   10   11
    6    7    9   12
    1    4   14   15

ans =

    1    3    9   11
    2    4   10   12
    5    7   13   15
    6    8   14   16

```

<i>Name</i>	<i>Size</i>	<i>Bytes</i>	<i>Class</i>	<i>Attributes</i>
<i>a</i>	<i>4x4</i>	<i>128</i>	<i>double</i>	

Variables

```
y=uint32(10)
whos y
x=1.2134
whos x
y=cast(x,"single")
tf=isnan(A)
A=0./[-2 -1 0 1 2]
TF = isnan(A)
```

y =

uint32

10

<i>Name</i>	<i>Size</i>	<i>Bytes</i>	<i>Class</i>	<i>Attributes</i>
<i>y</i>	<i>1x1</i>	<i>4</i>	<i>uint32</i>	

x =

1.2134

<i>Name</i>	<i>Size</i>	<i>Bytes</i>	<i>Class</i>	<i>Attributes</i>
<i>x</i>	<i>1x1</i>	<i>8</i>	<i>double</i>	

y =

single

1.2134

tf =

1x10 logical array

0 0 0 0 0 0 0 0 0 0

A =

0 0 NaN 0 0

TF =

1×5 logical array

0 0 1 0 0

String

```
str="Navneet has scored 9.0 CGPA"
str=unique(str)
str2=" Congratulations!!!"
str+str2
str=split(str+str2)
x=["1","2","3"
   "4","5","6"]
join(x,"+")
d="dffsf"
whos d
```

str =

"Navneet has scored 9.0 CGPA"

str =

"Navneet has scored 9.0 CGPA"

str2 =

" Congratulations!!!"

ans =

"Navneet has scored 9.0 CGPA Congratulations!!!"

str =

6×1 string array

"Navneet"
"has"
"scored"
"9.0"

```
"CGPA"  
"Congratulations!!!"
```

```
x =
```

```
2×3 string array
```

```
"1"    "2"    "3"  
"4"    "5"    "6"
```

```
ans =
```

```
2×1 string array
```

```
"1+2+3"  
"4+5+6"
```

```
d =
```

```
"dffsf"
```

Name	Size	Bytes	Class	Attributes
d	1x1	166	string	

Find

```
s1=["Navneet is Pre-Final yr student","he is","LT509"]  
p1="Navneet"  
contains(s1,p1)  
% Replace  
replace(str,"Navneet","Someone")  
t=datetime('now','TimeZone','local','Format','d-MMM-y HH:mm:ss Z')
```

```
s1 =
```

```
1×3 string array
```

```
"Navneet is Pre-Fina..."    "he is"    "LT509"
```

```
p1 =
```

```
"Navneet"
```

```
ans =
```

1×3 logical array

1 0 0

ans =

6×1 string array

*"Someone"
"has"
"scored"
"9.0"
"CGPA"
"Congratulations!!!"*

t =

datetime

8-Jun-2026 15:58:51 +0530

Structures

```
feild='f';  
value={'xyz';  
      [10,20,30];  
      magic(5)};  
s=struct(feild,value)
```

s =

3×1 struct array with fields:

f

Functions

```
x=10  
y=20  
f=@(x,y) (x.^2 - y.^2)  
a=6
```

```
a=10;  
CS(a)
```

x =

```
10

y =

    20

f =

function_handle with value:

    @(x,y) (x.^2-y.^2)

a =

    6

y =

    100

ans =

    100
```

dictionary

```
Products=["Apple","Mango","ORANGES"]
prices=[100 200 300.5]
d=dictionary(Products,prices)
```

```
Products =

    1×3 string array

    "Apple"    "Mango"    "ORANGES"
```

```
prices =

    100.0000    200.0000    300.5000
```

```
d =

dictionary (string --> double) with 3 entries:
```

```
"Apple"    --> 100  
"Mango"    --> 200  
"ORANGES"  --> 300.5000
```

time Series

```
ts1 = timeseries([5 10 15 20 25])  
  
timeseries  
  
Common Properties:  
    Name: 'unnamed'  
    Time: [5x1 double]  
    TimeInfo: tsdata.timemetadata  
    Data: [1x1x5 double]  
    DataInfo: tsdata.datametadata
```

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